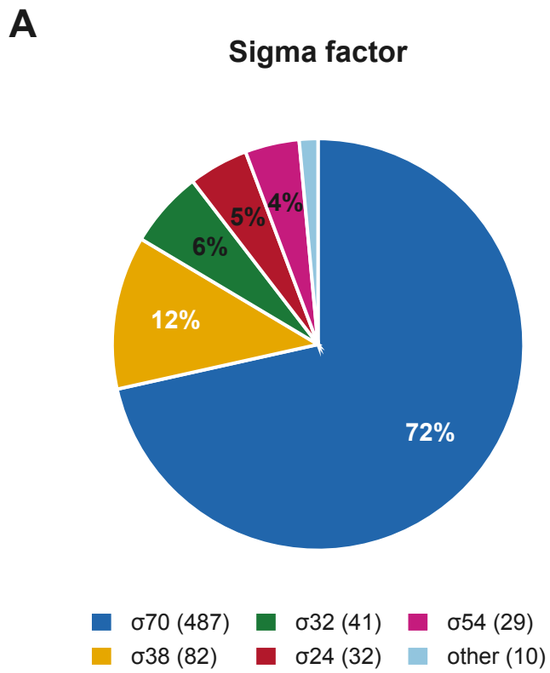
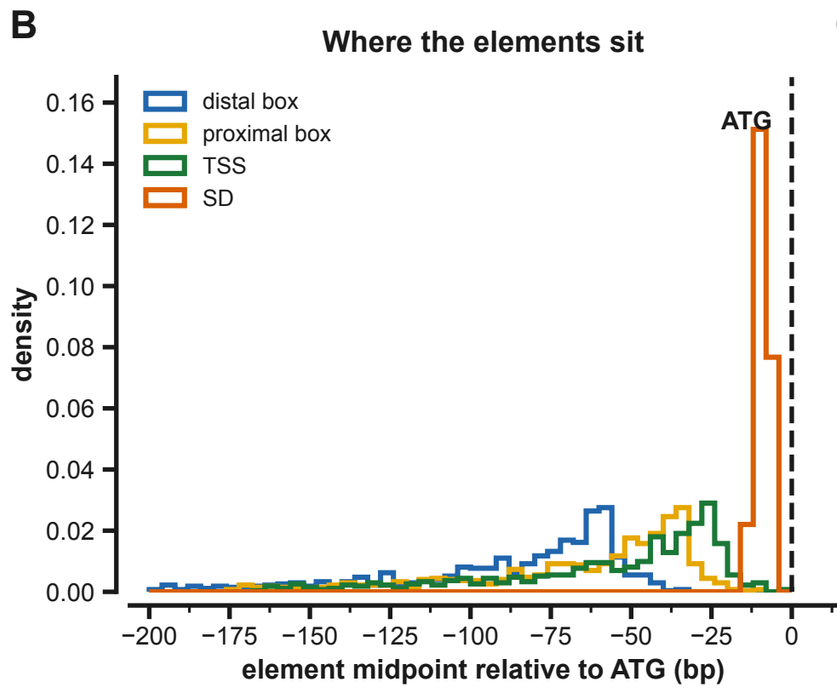
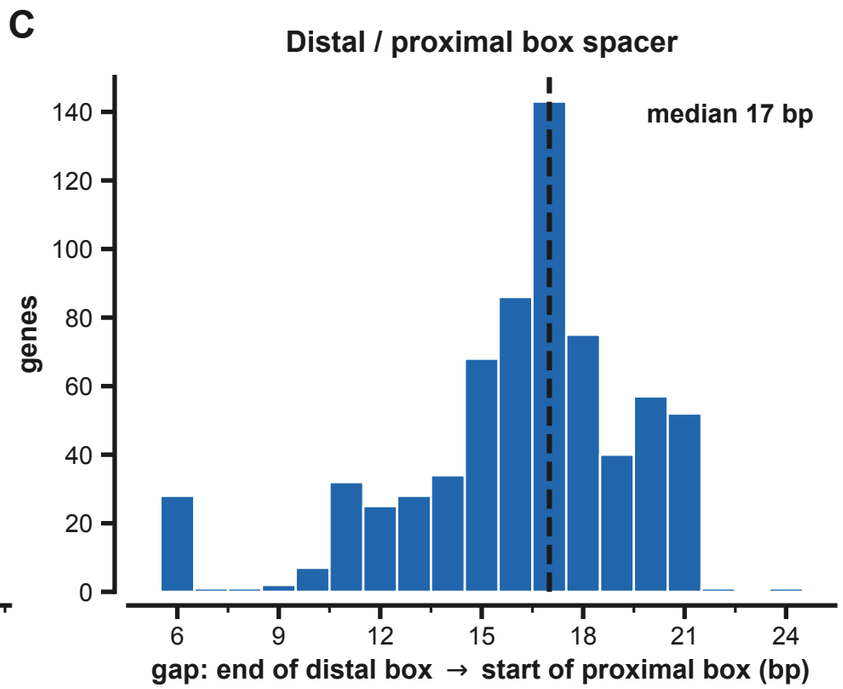
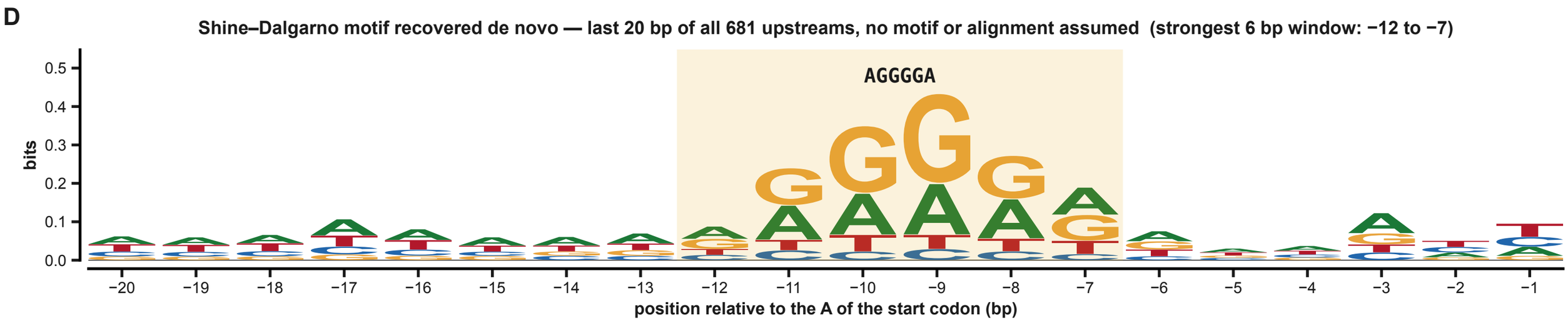
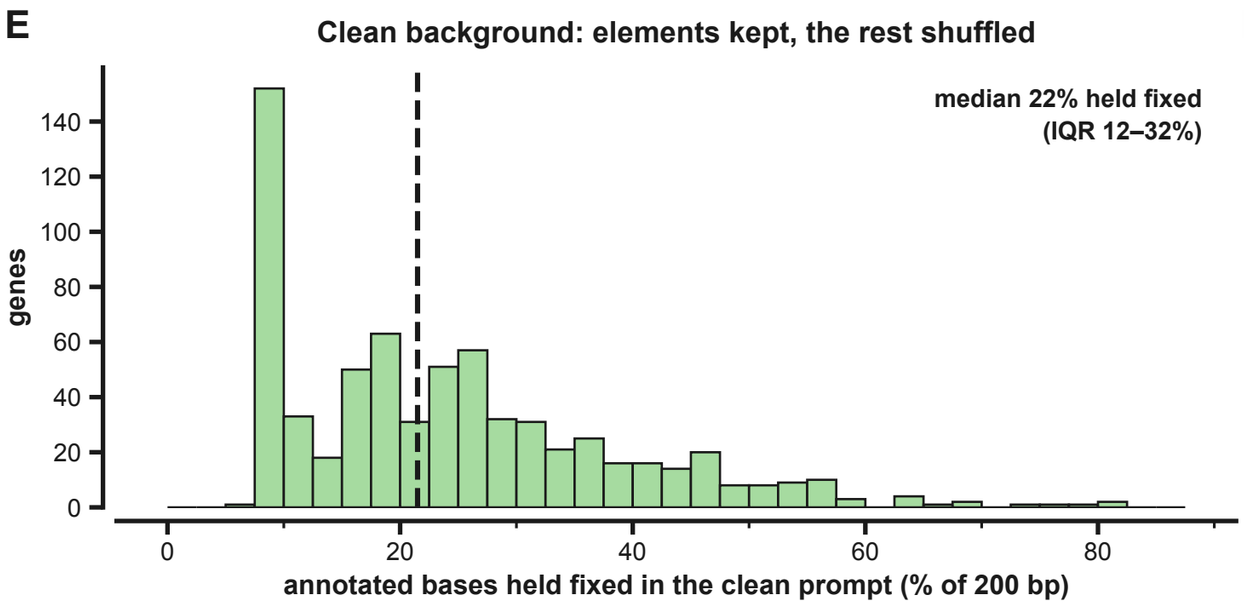
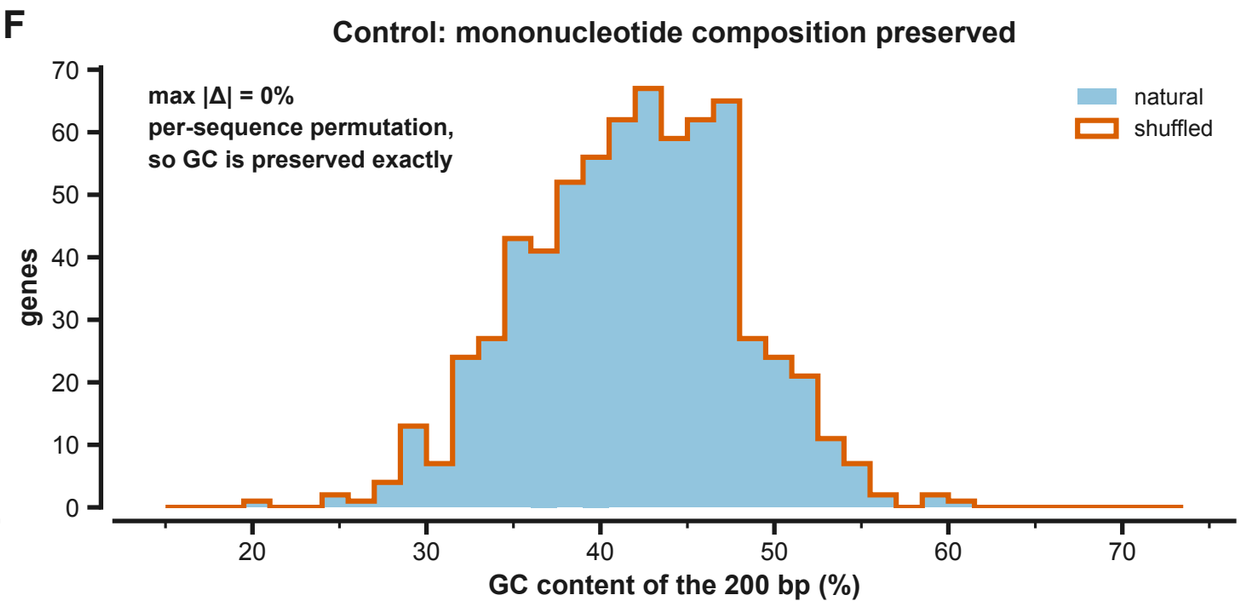

A
Sigma factor
72%
12%
6%
5%
4%
σ70 (487)
σ32 (41)
σ54 (29)
σ38 (82)
σ24 (32)
other (10)


B
Where the elements sit
distal box
proximal box
TSS
SD
ATG
density
element midpoint relative to ATG (bp)


C
Distal / proximal box spacer
median 17 bp
genes
gap: end of distal box → start of proximal box (bp)


D
Shine–Dalgarno motif recovered de novo — last 20 bp of all 681 upstreams, no motif or alignment assumed (strongest 6 bp window: −12 to −7)
AGGGGA
bits
position relative to the A of the start codon (bp)


E
Clean background: elements kept, the rest shuffled
median 22% held fixed
(IQR 12–32%)
genes
annotated bases held fixed in the clean prompt (% of 200 bp)


F
Control: mononucleotide composition preserved
max |Δ| = 0%
per-sequence permutation,
so GC is preserved exactly
natural
shuffled
genes
GC content of the 200 bp (%)